Case Studies in Thermal Engineering 42 (2023) 102741

Contents lists available at ScienceDirect

Case Studies in Thermal Engineering

journal homepage: www.elsevier.com/locate/csite

Performance enhancement of PV panels using phase change material (PCM): An experimental implementation

Hussein M. Maghrabie [a], A.S.A. Mohamed [b,c,*], Amany M. Fahmy [a], Ahmed A. Abdel Samee [a]

[a] Department of Mechanical Engineering, Faculty of Engineering, South Valley University, Qena, 8352, Egypt
[b] Mechanical Department, Faculty of Technology and Education, Sohag University, Sohag, 82524, Egypt
[c] High Institute for Engineering and Technology, Sohag, Egypt

ARTICLE INFO

Keywords:
PV panel
PCM thickness
Tilt angle
Thermal images
Temperature distribution
Electrical efficiency

ABSTRACT

In the present study, cooling of a photovoltaic (PV) panel using a phase change material (PCM) of paraffin wax RT-42 attached to a panel back surface is experimentally investigated. Two identical PV panels having a maximum electrical generated power of 40 W are employed to perform the outdoor experiments: a reference PV panel (PVr) and another integrated with PCM (PV-PCM). The current implementation is conducted with different thicknesses of PCM (1, 2, and 3 cm) varying the tilt angle of PV panels at 15°, 20°, 25°, and 30°. The infrared thermal camera is used to record the temperature distribution over the front surface of the PVr and PV-PCM panels. Results indicate that the highest temperature distribution appeared at the upper portion of the PV panels, whereas the lower part of the PV panels shows the lowest temperature distribution. At a tilt angle of 15°, the temperature at the panel's top side is higher than that of the bottom side by 17.1%, 15.7%, and 13.2% for PCM thickness of 1, 2, and, 3 cm, respectively. There is also enhancement in the electrical power output of 15.8% for the PV-PCM panel with a 3 cm PCM thickness over the reference panel (PVr) at a tilt angle of 30°. Relatedly, elevating the tilt angle from 15° to 30° improves the electrical efficiency of PV-PCM with a 3 cm thickness of the PCM by 7.4%. Finally, using PCM with a thickness of 3 cm enhances the electrical efficiency of the PV-PCM panel by 14.4% compared to that for the PVr at a tilt angle of 30°.

Nomenclature

A	panel surface area [m^2]
C	specific heat capacity [kJ/kg.k]
C_O	solid heat capacity [kJ/kg.k]
FF	fill factor
G	solar radiation intensity [W/m^2]
I	current [A]
k	thermal conductivity [W/m.K]
L	latent heat of fusion [kJ/kg]

* Corresponding author. Mechanical Department, Faculty of Technology and Education, Sohag University, Sohag, 82524, Egypt.
E-mail address: ahmedquse2000@yahoo.com (A.S.A. Mohamed).

https://doi.org/10.1016/j.csite.2023.102741
Received 20 August 2022; Received in revised form 23 October 2022; Accepted 14 January 2023
Available online 18 January 2023
2214-157X/Published by Elsevier Ltd. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

P	power [W]
P_S	saving power [W]
t	time [hr]
T	temperature [°C]
T_s	solid temperature [°C]
V	voltage [V]

1. Introduction

Due to the ever-increasing demand for energy in the economic, manufacturing, agricultural, and domestic sectors, fossil fuel supplies have become scarce. Subsequently, in recent years, renewable energy choices such as solar, wind, hydro, geothermal, bio-fuel, etc. have increasingly been used to produce electricity to support the sustainable development goals (SDGs) and mitigate the drastic climate change and the global warming [1–3]. Massive advances in photovoltaic systems for residential as well as commercial purposes are introduced to minimize the overall national energy requirements and reduce greenhouse gas emissions [4–6]. The solar photovoltaic (PV) panel is considered one of the most essential renewable energy sources for different applications [7,8].

Several parameters affect the operation efficiency of solar PV panels such as the incident solar radiation intensity and the working temperature of semiconductor cells. The heat absorbed increases the module's temperature, which adversely affects its output and consequently, the PV panel performance decreases as the surface temperature rises [9–11]. The relative efficiency reduction is about 0.45% with an increase of 1 °C over the designed operating temperature of 25 °C. To increase the efficiency of PV panel, it is necessary to decrease the PV temperature using active, passive, or hybrid cooling techniques [12,13].

Insulating the back surface of the PV panel results in increasing the PV temperature by around 12–20 °C and can decrease the electrical efficiency about 7–10% [14]. Wilson and colleagues [15] introduced an experimental work to investigate the impact of water flow on the backside of the PV panel. Results showed that the surface temperature of PV decreased quickly by almost 32 °C from 62 to 30 °C, leading to improving the PV performance by almost 13%. Maghrabie et al. [16] implemented the performance improvement of PV panels experimentally and found that using an air cooling system reduced the front surface temperature of PV cells by about 10% and the back surface by 11%.

Greek symbols

ρ	density [kg/m^3]
η	efficiency [%]
β	Volumetric expansion coefficient
θ	tilt angle [°]

Abbreviation

PV-PCM	PV panel with phase change material
PVr	PV panel reference

Subscripts

amb	ambient
e	enhancement
ele	electrical
inc	incident
l	liquid
max	maximum
oc	open circuit
sc	short circuit
s	solid

Farhana et al. [17] fabricated an active cooling PV system to decrease its temperature employing a brushless fan with an intake and exhaust manifold for uniform distribution of airflow. Results revealed that using a cooling system decreased the temperature of the solar PV panel to the maximum value of 12 °C. Teo et al. [18] studied the influence of employing copper fins attached to the back side of the PV panel using an air push fan. It was observed that the surface temperature of PV was reduced by around 30 °C. Fossa et al. [19] studied experimentally the mechanisms of heat transfer in a double-skin photovoltaic facade. Fossa et al. [19] studied experimentally the mechanisms of heat transfer in a double-skin photovoltaic facade. Results indicated that the increase in the channel spacing reduced the working temperature for all examined heating modes. Moreover, based on the obtained results, correlations for local and average Nusselt numbers were introduced.

Utilizing PCM integrated different thermal systems as passive cooling systems for appropriate thermal management [20–23]. A combined method of CFD numerical simulation, ANN, NSGA algorithm, and MCDM methods was introduced to maximize the stored energy of PCM [24]. Other scholars provided reliable models to predict the melting rate and the surface-averaged Nusselt number for the melting process of PCM in spherical thermal storage systems [24,25]. Bouzennada et al. [26] studied the heat distribution and the

melting time of a PCM capsule considering the position effect of the inner tube and its temperature wall numerically. Results revealed that both melting rates and heat transfer were enhanced when the inner tube moved downward or by raising the Stefan number.

Many experiments have been recently conducted to decrease the surface temperature of PV panels and regulate the thermal energy based on phase change materials [27]. Selecting the suitable material of PCM may lead to storing a large quantity of heat/cold by melting/solidification. The melting of PCM is characterized by a small change in volume typically less than 10%. Pichandi et al. [28] investigated experimentally the PV performance integrated with PCM. Results showed that an improvement of 1.21% in the daily average performance of the PV-PCM panel at an effective temperature reduction of 7 °C. Hasan and colleagues [29] investigated the performance of PV-PCM (RT-42) panel in a hot climate connected to the back surface of the PV panel numerically and experimentally. The PV-PCM panel ultimately obtained a 10.5 °C decrease in the average temperature of the PV panel at the peak time as well as the power output was increased yearly by 5.9%.

Indartono et al. [30] investigated the best thickness of PCM to reduce the temperature of PV panel using coconut and oil crude palm oil (CPO) as PCMs numerically and experimentally. Results showed that the 80 mm was the optimum thickness of PCM and the CPO had a better performance than the coconut oil in an ambient temperature around 27–30 °C. Cellura et al. [31] designed a PV-PCM panel using different types of paraffin wax and found that the maximum reduction in temperature was 22 °C. Savvakis et al. [32] conducted an experimental study to reduce and control the temperature of PV panels to enhance its performance. As a result, the PV with different phase change materials of PCM-27 and PCM-31 generated more energy by 4.19 and 4.24%, respectively, while the increase in PV efficiency due to PCM varied between 2.86 and 4.19%.

Khanna et al. [33] mathematically implemented an efficiency analysis of the tilt of PV-PCM using RT-25 HC. It was found that the PV temperature decreased from 43.4 °C to 34.5 °C as the tilt angle raised from 0° to 90° with an improvement in PV performance from 18.1% to 19%. Waqas et al. [34] introduced an experimental study on PV-PCM panels using RT-24 encapsulated in copper tubes. Results showed that the best reduction in temperature at the PV panel was 8.5 °C for 2.5 kg of PCM per unit of PV panel area and the efficiency was enhanced by up to 3%. In addition, Kumar et al. [35] experimentally studied the influence of using a combination of Paraffin Wax (Petroleum wax), Copper (Cu), and Silicon Carbide (Sic) on a PV panel's thermal actions and electrical performance. The electrical output generated from the PV panels increased by 2.8% when using the standard case and by 4.3% when using combined PCM. In addition, Khanna et al. [36] conducted a numerical implementation to find the most suitable depth of a PCM container and the optimum of a finned-PV-PCM for different solar irradiance levels. The results revealed that the minimum temperature was maintained at a 25 cm spacing between successive fins with a thickness of 2 mm.

Huang et al. [37] carried out an experimental investigation using three PV-PCM panels and two types of PCM. Results revealed that the RT-25 maintained the temperature under 29 °C for 130 min over the front surface of the PV panel. The thermal control was not effective when using Granular GR-40 as RT-25. Preet et al. [38] carried out an experimental implementation on water-based PV/T and water-based PV/T-PCM panels. It was observed that the maximum improvements in electrical efficiency were 10.66% and 12.6% for PV/T and PV/T-PCM, respectively. Browne et al. [39] carried out an experimental study on a novel PV/T/PCM panel using capric-palmitic fatty acid eutectic of PCM. Results indicated that the thermal efficiency of the PV/T-PCM panel ranged from 20% to 25%.

Nanoparticles are integrated with PCM to improve the thermophysical properties of pure PCM [40–43]. Inserting metal-oxide such as CuO, Al_2O_3, and TiO_2 nanoparticles and carbon-based nanomaterials such as MWCNT, GNP, and SWCNT was implemented to enhance the thermophysical properties of pure PCM [44]. Abdelrahman et al. [45] fabricated an experimental investigation using RT-35 HC PCM mixed with nanoparticles. Results indicated that adding nanoparticles to PCM reduced the temperature to 52.3% by changing the heat flux from 279 to 820 W/m^2 at T_{amb} of 20 °C. Abdelrazik et al. [46] conducted a numerical study on the PV/PCM and PV/T systems to study the impacts of nano-enhanced phase change materials on both thermal and electrical efficiency. Compared to a standalone PV system, the PV/T/nano-PCM system with 10% nanoparticles had a higher efficiency of 6.9% and 22%, respectively. Yang et al. [47] verified experimentally that the incorporation of the PCM layer into a PV/T panel successfully decreased its ambient heat loss and improved the system's primary energy-saving by around 14%.

Huang et al. [48] investigated the heat transfer enhancement in PCM using metallic fins to improve the heat transfer surface area and enhance the thermal homogeneity of PCM experimentally and theoretically. Nijmeh et al. [49] conducted an experimental analysis of the technical-economic benefits of using BioPCM to improve the power output of PV panel. The PV/BioPCM system had a yearly conversion efficiency of 12.5% compared to 12.08% for the standard PV system. Sarafraz et al. [50] investigated the thermal and electrical performance of a PV panel cooled with nano-PCM and a 50:50 of multi-walled carbon nanotube–water/ethylene glycol flowing into pipes. Results indicated that the MWCNT/WEG50 nano-suspension with a concentration of 0.2 wt% accomplished the best electrical and thermal performance of PV panel.

According to the brief previous available literature review, the accomplished studies have been concerned with investigating the enhancement of PV panels utilizing the PCM regardless of its melting characteristics. Therefore, the present study aims at studying the performance of a PV panel combined with paraffin wax RT-42 PCM and comparing it with a conventional PV panel as a reference panel (PVr). Relatedly, in the present implementation, the temperature distributions over the front surface of PVr and PV-PCM panels due to the melting characteristics of PCM are analyzed in order to study the uniformity of surface temperature using the infrared thermal camera. Furthermore, the short circuit current (I_{SC}), electrical power output (P_{ele}), and open-circuit voltage (V_{OC}) of the PVr and PV-PCM panels are recorded to assess the performance of PV-PCM panel and compare it with that for PVr panel. Lastly, the effects of PCM thickness (1, 2, and 3 cm) and the tilt angle of PV panels (15°, 20°, 25°, and 30°) on the electrical generated power and electrical efficiency are evaluated.

2. Methodology

With the aid of historical weather information, the methodology of the present study consists of experimental assessment of PCM performance for PV cooling in the hot climate of Qena, Egypt. Using fill factor (*FF*) on the estimated short circuit current (I_{SC}) and open-circuit voltage (V_{OC}) of the respective PV panels, and hence the electrical power is determined.

3. Experimental work

3.1. Experimental set-up

The real photo of the experimental set-up and the schematic diagram are presented in detail in Fig. 1. The experimental set-up is construed at the Faculty of Engineering, South Valley University, Qena, Egypt. Two identical polycrystalline silicon PV panels with a rated power capacity of 40 W and dimensions of 55 cm × 65 cm have been mounted for testing, one without PCM as a reference panel (PVr) and another with PCM (RT-42) and denoted by PV-PCM. The specifications of examined PV panels are presented in Table 1. The experimental set-up includes an adjustable stand, two photovoltaic polycrystalline silicon panels and one PCM-filled container with three different thicknesses of 1, 2, and 3 cm, and the measuring instruments. To regulate the tilt angle of PV systems, an adjustable stand is used and it is designed from rods with metal angles as one movable side to change the tilt angle of the PV panels. The stand is positioned to have the panels face south, and the tilt angle is set to be 15°, 20°, 25°, and 30° to the horizontal axis. The impact of the tilt angle of PV panels on the electrical efficiency as well as on the melting of PCM during the tests is investigated.

To accommodate the PV panels, a rectangular PCM container with inside dimensions of 55 cm × 65 cm x 4 cm made of a 2 mm thick galvanized sheet is used. The container is mounted to the back side of the PV panel and coated with a silicon adhesion to prevent the possible leakage when the PCM is completely melted. To enhance the performance of the solar panels experimentally and verify the feasibility of using PCM, the surrounding air temperature, incident solar radiation intensity, and voltage and current are measured. A Pyranometer is used to measure the incident solar irradiation. A Digital Multimeter is used for the voltage and current readings to calculate the V_{OC} and I_{SC} in the PV panels circuit for the reference PVr panel as well as for the PV-PCM. Additionally, an infrared camera is utilized to study the temperature distribution on the front surface of the PV panels at different sections, as shown in Fig. 2. The temperatures have been recorded for a large number of points on the panel surface. Therefore, symbols from T1 to T6 have been used, which express the average temperatures at the points indicated in Fig. 2. This is for more understanding for the figures in the results section.

At the latitude of 26.16° N, 32.71° E facing the south direction, the experiments are tested using Polycrystalline transparent epoxy resin PV panels. The electrical power output of PV panels is verified prior to the inclusion of the PCM in the panels by testing the V_{OC} and I_{SC}. As a reference, one PV panel is handled and another is fixed to galvanized sheet container filled with solid PCM and covered with a flat plate fabricated from a 1.5 mm thick sheet of aluminium alloy (1050A) and PV panel is glued by epoxy resin to the flat plate and container. The thermal and physical properties of employed RT-42 PCM are presented in Table 2. The experiments are carried out under real meteorological outdoor conditions.

3.2. Technique and data reduction

In carrying out the experimental work, the following techniques are used:

- Experimental tests are performed outdoor during the summer season.
- To investigate the effects of PCM with different thicknesses and with various tilt angles of the PV panels on the performance of integrated PV-PCM systems, two different PV panels are tested: one is a reference and the second with pure PCM.
- Short circuit current (I_{SC}) and open-circuit voltage (V_{OC}) are measured to evaluate the electrical power generated by PV panels.



1- PV panel. 2- PV-PCM panel. 3- Infrared camera. 4- Multimeter. 5- Pyranometer.

Fig. 1. Real photo of the experimental set-up of PVr and PV-PCM panels.

Table 1
Specifications of the examined PV panels.

Parameter	Value
Model	polycrystalline silicon
Max. power, P_{max} [W]	40
Short circuit current, I_{SC} [A]	4.3
Open-circuit voltage, V_{OC} [V]	12
Fill factor, FF [−]	0.78
Electrical efficiency, η_{ele} [%]	16.5
Operation temperature, T [°C]	0–70
Standard testing condition	25 °C and 1000 W/m^2
Dimensions	55 × 65 × 2 cm^3

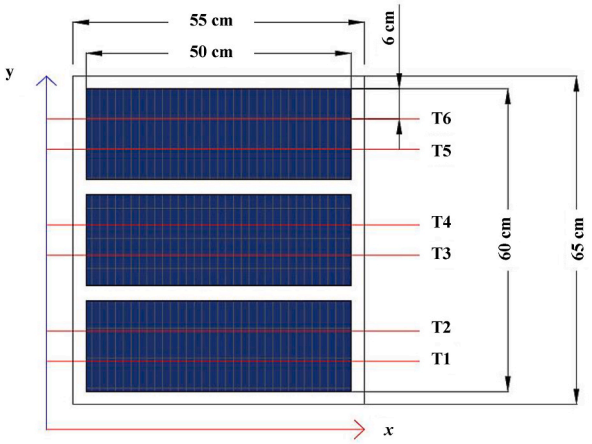


Fig. 2. Schematic of PV panel showing the temperature distribution.

Table 2
Thermal and physical properties of RT-42 PCM.

Parameter	Value
Chemical classification	Paraffin wax
Melting temperature, T [°C]	38–43
Specific heat capacity, C [kJ/kg.K]	2
Density solid, ρ_s [kg/m^3]	880
Density liquid, ρ_l [kg/m^3]	760
Thermal conductivity, k [W/m.K]	0.2
Volume expansion, [%]	12.5
Maximum operation temperature, T_{max} [°C]	72

- The temperature distribution on the front surface of the reference PVr and PV-PCM panels is measured and recorded using an infrared camera.
- To evaluate the effect of using PCM material with different thicknesses, the enhancement in producing the electrical power due to employing PV-PCM panel is assessed.

The data are gathered every 30 min, from 10:00 to 16:00 and the solar radiation intensity, thermal photos, V_{OC}, and I_{SC} for PV panels are reported for each experiment. To obtain the performance of photovoltaic panels under the test conditions, the following procedures are followed.

1. Cleaning both PV panels with dry cotton before each test to avoid the accumulation of dust.
2. Ensuring that the electric circuit is equipped for measurements of voltage and current to calculate the electrical power output and to estimate the maximum power (P_{max}) to draw the characteristics I–V curve for the two PV panels to estimate the electrical efficiency.
3. Using the digital solar power Pyranometer to record the solar radiation intensity (G) and with the aid of the soda-pro website to determine the air temperature and relative humidity.
4. Setting the Multimeter to the appropriate position to measure the I_{SC} and V_{OC}.
5. Taking a shot of an infrared camera for two PV panels every 30 min to study the temperature distribution on the front surface of the PV panels.

For both panels, the voltage, current, and power are studied and analyzed to draw the characteristic $I-V$ curve as shown in Fig. 3. From the following equations, the PV solar panel's efficiency and saving power are determined [51]. The incident power over the PV panels is given by:

$$P_{inc} = G.A \tag{1}$$

where G is the incident solar radiation intensity, [W/m^2] and A is the panel area, [m^2].

In addition, using the measured values of voltage (V) and current (I) the PV panel electrical output power (P_{ele}) is calculated by:

$$P_{ele} = V.I \tag{2}$$

Also, the maximum obtained power (P_{max}) is calculated by the following equation:

$$P_{max} = V_{max}.I_{max} \tag{3}$$

where V_{max} and I_{max} are the maximum values of voltage and current.

The V_{OC} and I_{SC} are the maximum voltage and current from PV panels, respectively. However, at these operating conditions, the power from PV panels is zero. The fill factor (FF) is the parameter that conjuncts the V_{OC} and I_{SC}, determines the maximum value of power from PV panel. The FF is defined as the ratio between the maximum power from the PV panel and the multiplication of V_{OC} and I_{SC} as following:

$$FF = \frac{P_{max}}{V_{oc}.I_{sc}} \tag{4}$$

The most common equation used to estimate the electrical efficiency (η_{ele}) of PV panel is shown below:

$$\eta_{ele} = \frac{P_{max}}{P_{Inc}} \tag{5}$$

3.3. Uncertainty of experimental data

The measured parameters in this study required an accurate assessment to judge their accuracy. Therefore, the uncertainty ($u_{\bar{x}}$) in each variable (x') is described by specifying the best value ($\bar{x}$), for the variable followed by the uncertainty interval [52].

$$x' = \bar{x} + u_x \tag{6}$$

where x' is the measured variable, $\bar{x}$ is its best value, and u_x is the uncertainty.

The uncertainties in the present experimental measurements employed to assess the solar PV panel's performance are carried out using the method presented by Ref. [53]. The accuracy in each measured variables such as V, I, and G has a great effect on the uncertainty of the calculated values of PV such as the electrical output power and the electrical efficiency (η_{ele}). Thus, the uncertainty (u_η) of the calculated electrical efficiency (η_{ele}), due to uncertainties of individual variables were evaluated using the least-square fit analysis as shown below:

$$u_x = \left[\left(\frac{\partial x}{\partial x_1}\delta x_1\right)^2 + \left(\frac{\partial x}{\partial x_2}\delta x_2\right)^2 + \ldots + \left(\frac{\partial x}{\partial x_n}\delta x_n\right)^2\right]^{\frac{1}{2}} \tag{7}$$

The accuracy and maximum uncertainties of the experimental instruments are listed in Table 3. The uncertainty determined for the

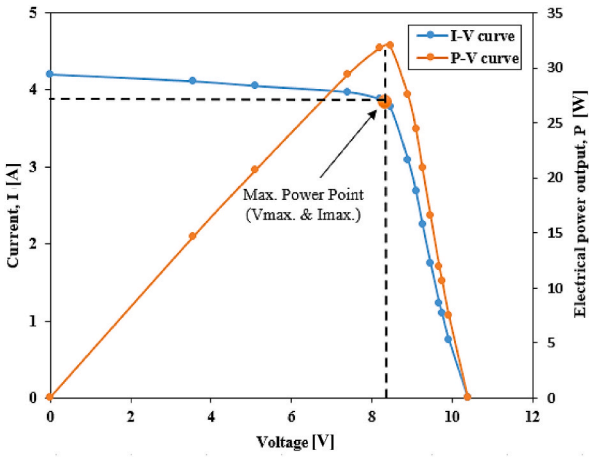


Fig. 3. Characteristic curve of PV panel.

Table 3
Uncertainty values in measurements and calculations.

Parameters	Maximum Uncertainty
Temperature, T	±0.17%
Voltage, V	±0.06%
Current, I	±0.14%
Incident radiation, G	±0.8%
Electrical efficiency, η_{ele}	±3.24%
Tilt angle, θ	±1.15%

electrical output power is 4.18%.

4. Results and discussion

In the present study, the effect of utilizing PCM (RT-42) for cooling PV panel on its performance is experimentally implemented. During the experiments, the effect of using PCM connected to the back surface of the PV-PCM panel with different thickness and four tilt angles on the performance of PV Panel are investigated. Two similar PV panels with the maximum power capacity of 40 W are examined and installed outdoor facing the south direction. One of the PV panels is directly attached to a PCM container with different thicknesses of 1, 2, and 3 cm named PV-PCM and another cooled with ambient air is the reference panel indicated by PVr. The two panels are studied at the same time under the same climatic conditions. The PV panels are holed using a movable stand to be able to change the tilt angle of PV panels at 15°, 20°, 25°, and 30°. The temperature distributions (T) on the front surfaces of the PV panels, short circuit current (I_{SC}), open-circuit voltage (V_{OC}), electrical power output (P_{ele}), and electrical efficiency (η_{ele}) of PV panels are recorded and analyzed.

4.1. Meteorological data

The performance of PV panels depends mainly on a variety of climatic variables such as solar radiation intensity and ambient air temperature. To study the performance of PV panels, the recorded results of these two parameters are therefore analyzed. Fig. 4 shows the variation of average solar radiation intensity and average ambient air temperature during the period of experiments. It is observed that the solar radiation intensity increases from 669 W/m^2 at 10:00 to reach its maximum value of 809 W/m^2 at 12:00 and then it tends to decreases until 296 W/m^2 at 16:00. Moreover, the average ambient temperature is approximately 32 °C at 10:00 and starts to increase until it reaches 34–36 °C about noon and then reaches to 39 °C at 16:00.

4.2. Temperature distribution on PV panels

The thermal images of the PV panels during the experiments are captured using an infrared camera type IC 120 LV TROTEC. The emissivity value for Poly-crystalline PV panels is assumed as 0.9 to achieve the right measurement of the images. Fig. 5 shows the thermal images for the temperature distribution over the front surface of the two panels PVr and PV-PCM panels for different PCM thicknesses of 1 cm, 2 cm, and 3 cm at incident solar radiation intensity of 800 W/m^2 and 30° tilt angle. The obtained images show considerable differences in the temperature distribution over the front surface between the two panels PVr panel and the PV-PCM panel. The temperature distribution over the front surface of the PVr panel is higher than the PV-PCM panel. In addition, the PV-PCM panel with a thickness of 3 cm indicates a temperature distribution less than those for the 2 cm and 1 cm thickness.

Furthermore, it is observed that for a reference PVr panel, the temperature distributions are substantially the same over the front surface, while for the PV-PCM panel, there is a clear difference between the temperature distributions along the panel surface. In the

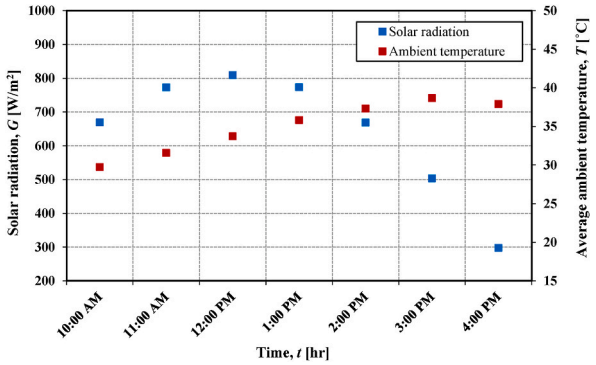


Fig. 4. Average solar radiation and average ambient temperature recorded in Qena, Egypt.

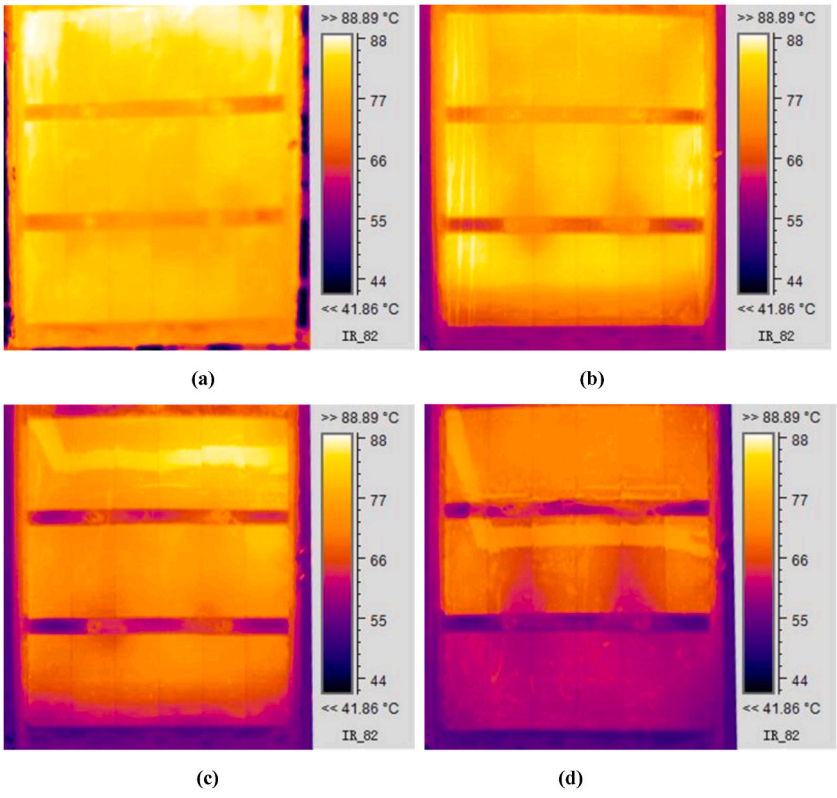


Fig. 5. Infrared camera thermal images at tilt angle of 30° and 800 W/m2 for (a) PVr and PV- PCM with PCM thicknesses of (b) 1 cm, (c) 2 cm, and (d) 3 cm.

PV-PCM panel, the highest temperature distribution appears at the upper portion of the panel, whereas the lower part shows the lowest temperature distribution. This is likely due to the fact that upon the initiation of the melting of PCM, it starts to move downwards due to its weight and the gravity force effect, which leads to leaving an empty part at the top of the cell that does not contain a PCM material. In addition, it is noted that the chosen temperature scale 44–88 °C and the color scheme, the PV panel's calculated temperature values vary within the 50–75 °C, while the ambient temperature is nearly 39 °C during the experiments.

Fig. 6 illustrates the influence of PCM thicknesses (1, 2, and, 3 cm) on the temperature distribution on the front surface of PV-PCM panel with different tilt angles of 15°, 20°, 25°, and, 30° at solar radiation intensity of 800 W/m^2. It is observed that for all thicknesses of the PCM, increasing the tilt angle of the panel reduces the surface temperature. When the PV-PCM panel is installed at an angle of 15°, it records the highest temperature distribution over the panel surface, while the lowest temperature is recorded for the surface of the solar panel at a tilt angle of 30°. This is likely ascribed to the fact that the energy flow within the PCM due to the convection is very low when the tilt angle is small [54]. Moreover, results show that at all values of tilt angles, increasing PCM thickness reduces the increment of the panel surface temperatures from T1 to T6. At a tilt angle of 15°, the temperature at the top side of PV panel (T6) is greater than that of the bottom side (T1) by 17.1%, 15.7%, and 13.2% for PCM thickness of 1, 2, and, 3 cm, respectively.

Fig. 7 illustrates the impact of increasing the PCM thickness on the temperature distribution on the front surface of the PV-PCM panel at 800 W/m^2 at different tilt angles of 15°, 20°, 25°, and 30°. The experimental results reveal that increasing the PCM thickness on the back surface of the PV panel decreases significantly the surface temperature of PV panels. Furthermore, due to the melting characteristics of employing PCM for cooling the back side of the PV panel, the front surface temperature at the top side of PV panel (T6) is higher than that the temperature at the bottom side of the PV panel (T1) at 1 cm PCM thickness by 15.7%, 17.6%, 18.2%, and 19% for a tilt angle of 15°, 20°, 25°, and 30°, respectively.

The temperature distribution on the front surface of PV panel using PCM on the back side is affected significantly by the PCM thicknesses and the tilt angle of the PV panel. Using PCM on the back side of the PV panel in a container with one volume leads to a non-uniformity of temperature distribution on the front surface. This implies that the temperature of each cell is different, that has a great impact on the output of electrical energy as well as on the cell's life span and hence the lifetime of the PV panel.

4.3. Electrical power output

Fig. 8 shows the electrical power versus the solar radiation intensity for 3 cm PCM thickness for different tilt angles of PV panels (15°, 20°, 25°, and 30°). Experimental results reveal that the maximum value obtained of the electrical power output for the reference PVr panel is 31.03 W when the solar radiation intensity reaches 800 W/m^2, while the value of the PV-PCM panel is 33.47 W at the same solar radiation intensity when using 3 cm thickness of PCM at a tilt angel of 30°. This implies that decreasing the panel surface

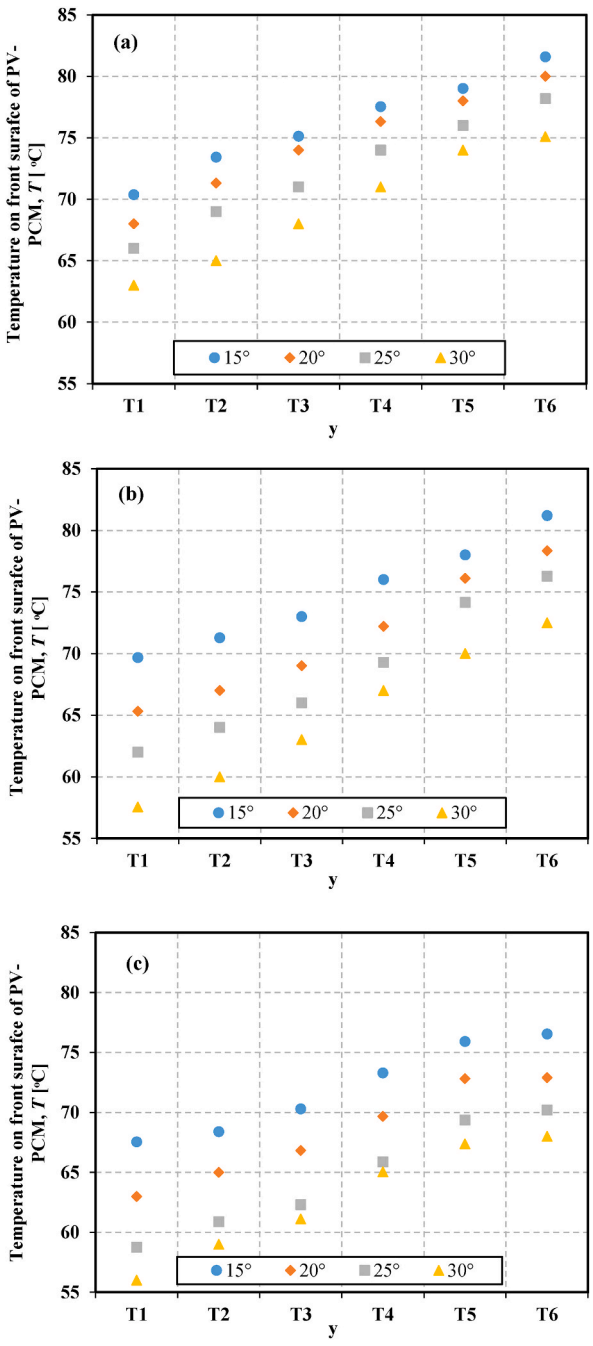


Fig. 6. Temperature distribution on the front surface of PV-PCM panel at different positions with varying tilt angel for PCM thickness of (a) 1 cm (b) 2 cm (c) 3 cm.

temperature increases the panel output power. Moreover, the enhancement in the electrical power output of the PV panels due to using PCM for all thicknesses greater than that for references panel (PVr) is more observed with increasing the solar radiation intensity.

4.4. Electrical efficiency

The electrical efficiency of PV panels is calculated as the ratio between the electrical output power supplied by PV panels and the solar radiation intensity. Since both the supplied power and the radiation received vary throughout the day with the same pattern, it is predicted that the performance of PV panels also varies throughout the day. Fig. 9 presents the variation of the electrical efficiency for the PVr panel and PV-PCM panel with 3 cm thickness of PCM panel versus the solar radiation at different tilt angles 15°, 20°, 25°, and 30°. It is observed that the efficiency of the PV-PCM panel is greater than that for PVr panel at all tilt angles. At a 30° tilt angle, results indicate that the maximum value of electrical efficiency of the PV panel is 13.1% at a solar radiation intensity of 800 W/m^2, while the

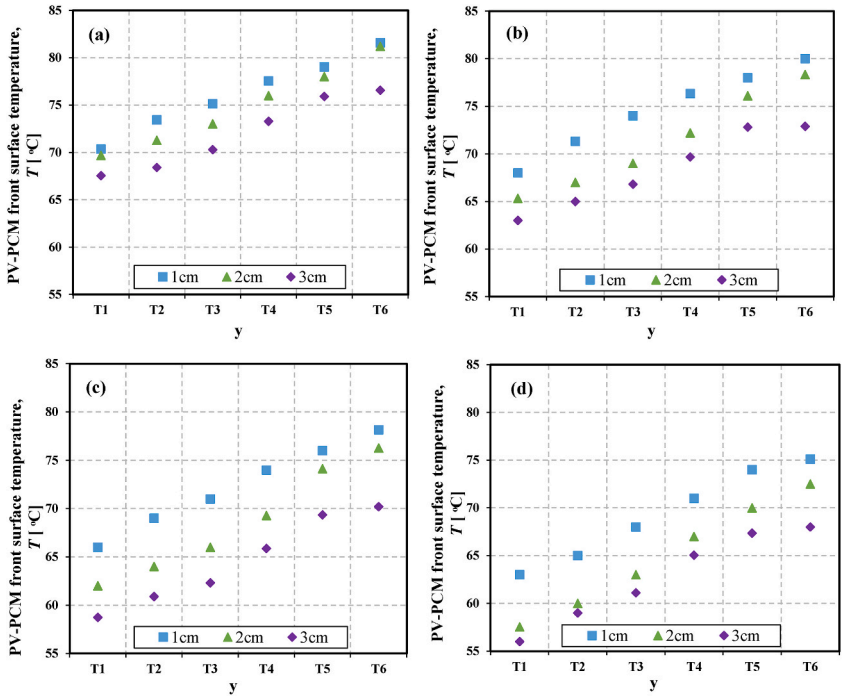


Fig. 7. Temperature distribution on the front surface of PV-PCM panel for different PCM thicknesses at tilt angle of (a) 15° (b) 20° (c) 25° (d) 30°.

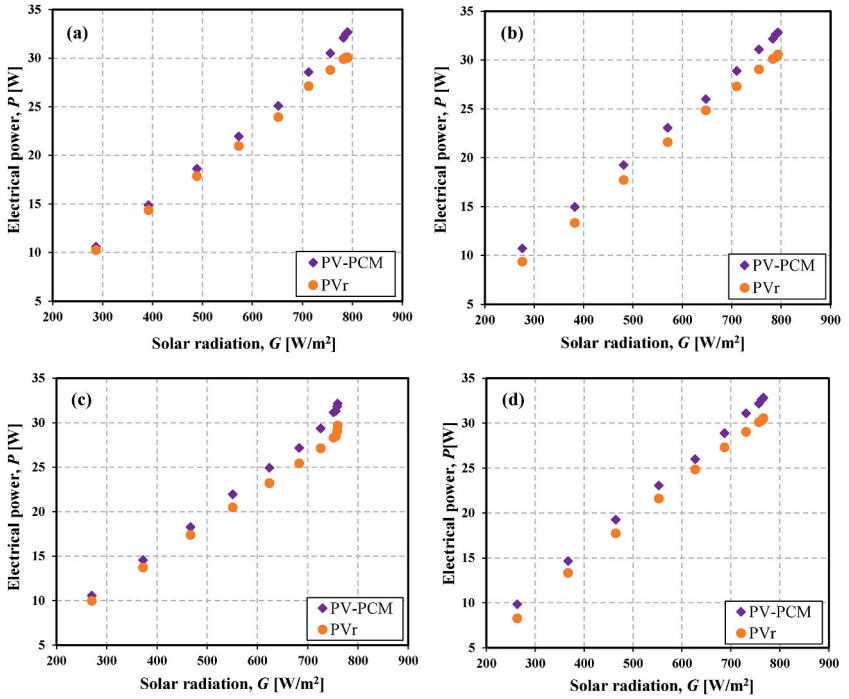


Fig. 8. Electrical power output for 3 cm PCM thickness at tilt angle of (a) 15° (b) 20° (c) 25° (d) 30°.

corresponding value for PV-PCM panel is 14.36% at the same value of solar radiation intensity. Moreover, for all values of the tilt angle of the PV panels, the enhancement in thermal efficiency due to using the PV-PCM is reduced with increasing the solar radiation intensity. In addition, the maximum value of the thermal efficiency for both PV panels is maintained at the 30° tilt angle.

Fig. 10 shows the average electrical power output and the average electrical efficiency variation at different tilt angles for PVr and

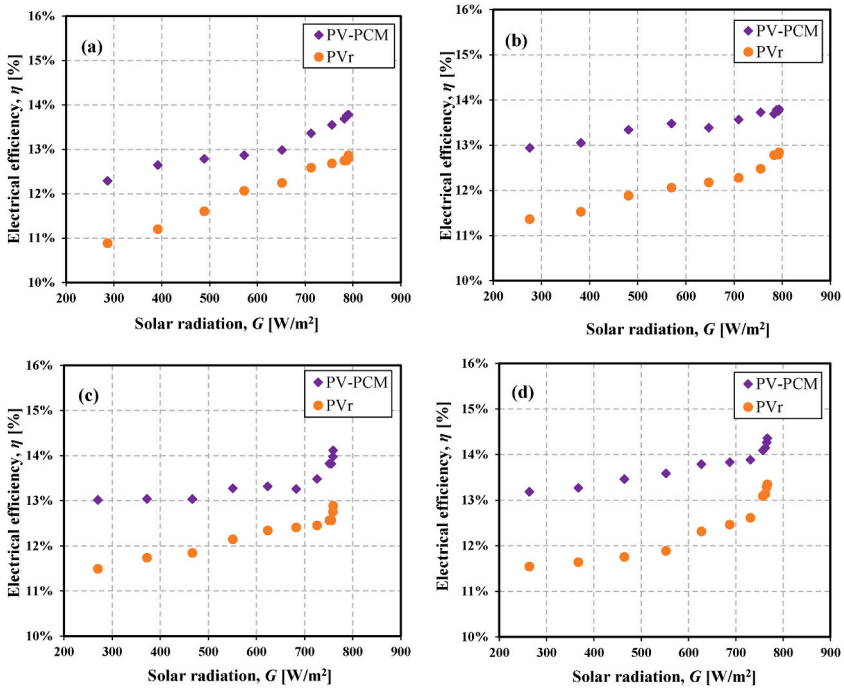


Fig. 9. Electrical efficiency versus solar radiation for PVr and PV-PCM panels with a 3 cm PCM thickness at tilt angle of (a) 15° (b) 20° (c) 25° (d) 30°.

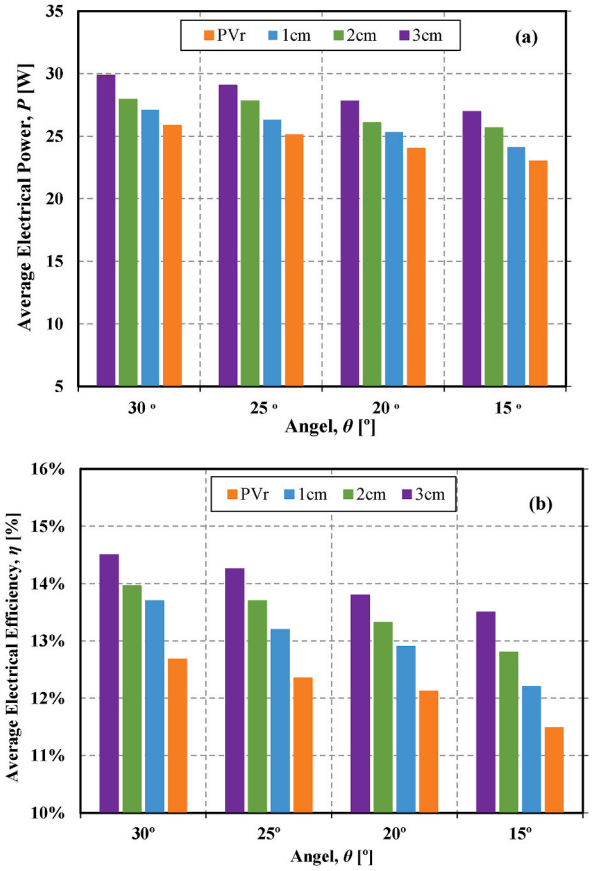


Fig. 10. (a) Average electrical power output (b) Average electrical efficiency for PVr and PV-PCM panels at different PCM thicknesses.

PV-PCM panels at different PCM thicknesses. At the 30° tilt angle of PV panels, the average power for the PVr panel is 25.9 W, while the corresponding value for the PV-PCM panel is 27.1 W, 28 W, and 30 W at PCM thicknesses of 1 cm, 2 cm, and 3 cm, respectively. Moreover, the corresponding values for the PVr panel are 23 W, 24 W, 25.1 W, and 25.9 W. However, those of the PV-PCM with 3 cm PCM are 27 W, 27.8 W, 29.1 W, and 30 W at a tilt angle of 15°, 20°, 25°, and 30°, respectively. These values reflect a significant improvement in power output of 15.8% for the PV-PCM panel with a 3 cm PCM compared to PVr panel at a tilt angle of 30°. Furthermore, the average electrical power output decreases with decreasing the tilt angle of the PV panels. In addition, experimental results show that the electrical efficiency is enhanced significantly with increasing the tilt angle as well as with increasing the thickness of PCM. Increasing the tilt angle from 15° to 30° increases the electrical efficiency of PV-PCM with a 3 cm thickness by 7.4%. Lastly, at a tilt angle of 30°, using PCM with a thickness of 3 cm enhances the efficiency of PV-PCM panel by 14.4% compared to the PVr panel.

5. Conclusions

In this study, the effect of PV panel cooling using PCM type RT-42 on the PV performance is experimentally implemented. Two identical PV panels are used and installed outdoor facing the south direction. One of the PV panels is directly attached to a PCM container with different thicknesses of 1 cm, 2 cm, and 3 cm called PV-PCM and another is the reference panel (PVr) cooled with the ambient surrounding air. The temperature distributions (T), short circuit current (I_{SC}), open-circuit voltage (V_{OC}), electrical power output (P_{ele}), and electrical efficiency (η_{ele}) of PV panels are recorded and analyzed. To conclude, the incorporation of PCM at the back side of the PV panel reduces the panel temperature distribution, leading to an improvement in the output power and the electrical efficiency. Using the infrared camera also gives deep details and explains the temperature distribution on the surface of the PV-PCM panel. Examining the infrared camera thermal images, it is found that the highest temperature is maintained in the upper part of the PV-PCM panel. Relatedly, at tilt angle of 15°, the temperature at the panel top side (T6) is greater than that of the bottom side (T1) by almost 17.1%, 15.7%, and 13.2% for PCM thickness of 1, 2, and, 3 cm, respectively. This confirms that the upper part has less cooling due to the melting of the PCM and sliding downwards, leaving a vacuum in the upper part of the PV-PCM panel.

The PV-PCM panel with a 3 cm PCM improves the power output compared to the PVr panel at a tilt angle of 30° by 15.8%. Additionally, the tilt angel of 30° has the best performance in all cases compared to 15°, 20°, and 25°. Furthermore, the electrical efficiency is improved significantly with increasing the tilt angle as well as with increasing the thickness of PCM. Changing the tilt angle from 15° to 30° enhances the electrical efficiency of PV-PCM with 3 cm thickness by 7.4%. Finally, using PCM with a thickness of 3 cm increases the electrical efficiency of PV-PCM panel by 14.4% compared to the PVr reference panel.

CRediT authorship contribution statement

Hussein M. Maghrabie: Conceptualization, Data curation, Methodology, Investigation, Formal analysis; Supervision, Writing - review & editing. **A. S. A. Mohamed:** Data curation, Methodology, Investigation, Formal analysis; Supervision, Writing - review & editing, Supervising on experimental measurements. **Amany M. Fahmy:** Collected Data, Designed and manufactured of the Experimental set-up, Perform experimental measurements, Writing the draft manuscript. **Ahmed A. Abdel Samee:** Data curation, Methodology, Investigation, Formal analysis; Supervision, Writing - review & editing, Supervising on experimental measurements.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

References

[1] A.G. Olabi, K. Obaideen, K. Elsaid, T. Wilberforce, E. Taha, H.M. Maghrabie, M. Ali, Assessment of the pre-combustion carbon capture contribution into sustainable development goals SDGs using novel indicators, Renew. Sustain. Energy Rev. 153 (2022), 111710, https://doi.org/10.1016/j.rser.2021.111710.
[2] K. Obaideen, M. Ali, T. Wilberforce, K. Elsaid, H.M. Maghrabie, A.G. Olabi, Biogas role in achievement of the sustainable development goals: evaluation, Challenges, and Guidelines, J. Taiwan Inst. Chem. Eng. 131 (2022), 104207, https://doi.org/10.1016/j.jtice.2022.104207.
[3] A.G. Olabi, T. Wilberforce, K. Elsaid, E.T. Sayed, H.M. Maghrabie, M.A. Abdelkareem, Large scale application of carbon capture to process industries – a review, J. Clean. Prod. 362 (2022), 132300, https://doi.org/10.1016/j.jclepro.2022.132300.
[4] H.M. Maghrabie, K. Elsaid, E.T. Sayed, M.A. Abdelkareem, T. Wilberforce, A.G. Olabi, Building-integrated photovoltaic/thermal (BIPVT) systems: applications and challenges, Sustain. Energy Technol. Assessments 45 (2021), 101151, https://doi.org/10.1016/j.seta.2021.101151.
[5] H.M. Maghrabie, M.A. Abdelkareem, A. Hai, A. Alami, M. Ramadan, E. Mushtaha, T. Wilberforce, A.G. Olabi, State-of-the-Art technologies for building-integrated photovoltaic systems, Buildings 11 (2021) 383.
[6] A.S.A. Mohamed, H.M. Maghrabie, Techno-economic feasibility analysis of Benban solar Park, Alex. Eng. J. 61 (2022) 12593–12607, https://doi.org/10.1016/j.aej.2022.06.034.
[7] T. Salameh, P.P. Kumar, A.G. Olabi, K. Obaideen, E. Taha, H.M. Maghrabie, M. Ali, Best battery storage technologies of solar photovoltaic systems for desalination plant using the results of multi optimization algorithms and sustainable development goals, J. Energy Storage 55 (2022), 105312, https://doi.org/10.1016/j.est.2022.105312.
[8] H.M. Maghrabie, M. Ali, K. Elsaid, E. Taha, A. Radwan, H. Rezk, T. Wilberforce, A.G. Abo-khalil, A.G. Olabi, A review of solar chimney for natural ventilation of residential and non-residential buildings, Sustain. Energy Technol. Assessments 52 (2022), 102082, https://doi.org/10.1016/j.seta.2022.102082.
[9] B. RamKiran, C. Sundarabalan, S. Kumarasamy, Sustainable passive cooling strategy for PV module: a comparative analysis, Case Stud. Therm. Eng. 27 (2021), 101317.

[10] M.S. Ahmed, A.S.A. Mohamed, H.M. Maghrabie, Performance evaluation of combined photovoltaic thermal water cooling system for hot climate regions, J. Sol. Energy Eng. 141 (2019), 041010, https://doi.org/10.1115/1.4042723.
[11] H.M. Maghrabie, A.S.A. Mohamed, M. Salem Ahmed, Experimental investigation of a combined photovoltaic thermal system via air cooling for summer weather of Egypt, J. Therm. Sci. Eng. Appl. 12 (2020) 1–9, https://doi.org/10.1115/1.4046597.
[12] A. Abdallah, R. Opoku, C.K.K. Sekyere, S. Boahen, K.O. Amoabeng, F. Uba, G.Y. Obeng, F.K. Forson, Experimental investigation of thermal management techniques for improving the efficiencies and levelized cost of energy of solar PV modules, Case Stud. Therm. Eng. 35 (2022), 102133.
[13] M. Omri, F. Selimefendigil, H.T. Smaoui, L. Kolsi, Cooling system design for photovoltaic thermal management by using multiple porous deflectors and nanofluid, Case Stud. Therm. Eng. 39 (2022), 102405.
[14] M.J.M. Pathak, K. Girotra, S.J. Harrison, J.M. Pearce, The effect of hybrid photovoltaic thermal device operating conditions on intrinsic layer thickness optimization of hydrogenated amorphous silicon solar cells, Sol. Energy 86 (2012) 2673–2677.
[15] E. Wilson, Theoretical and operational thermal performance of a "wet" crystalline silicon PV module under Jamaican conditions, Renew. Energy 34 (2009) 1655–1660.
[16] H.M. Maghrabie, A.S.A. Mohamed, M.S. Ahmed, Improving performance of photovoltaic cells via active air cooling system, in: Proc. 4th Int. Conf. Energy Eng., Aswan University, Egypt, 2017, pp. 1–5.
[17] Z. Farhana, Y.M. Irwan, R.M.N. Azimmi, A.R.N. Razliana, N. Gomesh, Experimental investigation of photovoltaic modules cooling system, in: 2012 IEEE Symp. Comput. Informatics, 2012, pp. 165–169. Penang.
[18] H.G. Teo, P.S. Lee, M.N.A. Hawlader, An active cooling system for photovoltaic modules, Appl. Energy 90 (2012) 309–315, https://doi.org/10.1016/j.apenergy.2011.01.017.
[19] M. Fossa, C. Ménézo, E. Leonardi, Experimental natural convection on vertical surfaces for building integrated photovoltaic (BIPV) applications, Exp. Therm. Fluid Sci. 32 (2008) 980–990.
[20] A.H. Alami, H.M. Maghrabie, M.A. Abdelkareem, E.T. Sayed, Z. Yasser, T. Salameh, S.M.A. Rahman, H. Rezk, A.G. Olabi, Potential applications of phase change materials for batteries' thermal management systems in electric vehicles, J. Energy Storage 54 (2022), 105204, https://doi.org/10.1016/j.est.2022.105204.
[21] J. Chiew, S.C. Chin, W.D. Toh, Z. Gao, J. Jia, Thermal state-of-expansion or melting of phase change material based heat sink for underwater battery power system, J. Energy Storage 26 (2019), 100956.
[22] M. Algarni, M. Alazwari, M.R. Safaei, Optimization of nano-additive characteristics to improve the efficiency of a shell and tube thermal energy storage system using a hybrid procedure: DOE, ANN, MCDM, MOO, and CFD modeling, Mathematics 9 (2021) 3235.
[23] J. Chiew, S.C. Chin, W.D. Toh, Z. Gao, J. Jia, Low-temperature macro-encapsulated phase change material based thermal energy storage system without air void space design, Appl. Therm. Eng. 141 (2018) 928–938.
[24] H. Maleki, M. Ashrafi, N.Z. Ilghani, M. Goodarzi, A. Muhammad, Pareto optimal design of a finned latent heat thermal energy storage unit using a novel hybrid technique, J. Energy Storage 44 (2021), 103310.
[25] M. Goodarzi, M.A. Elkotb, A.K. Alanazi, H.M. Abo-Dief, I.B. Mansir, V. Tirth, F. Gamaoun, Applying Bayesian Markov chain Monte Carlo (MCMC) modeling to predict the melting behavior of phase change materials, J. Energy Storage 45 (2022), 103570.
[26] T. Bouzennada, F. Mechighel, A. Filali, K. Ghachem, L. Kolsi, Numerical investigation of heat transfer and melting process in a PCM capsule: effects of inner tube position and Stefan number, Case Stud. Therm. Eng. 27 (2021), 101306.
[27] H.M. Maghrabie, A.S.A. Mohamed, A.M. Fahmy, A.A.A. Samee, Performance augmentation of PV panels using phase change material cooling technique: a review, SVU-International J. Eng. Sci. Appl. 2 (2021) 1–13, https://doi.org/10.21608/SVUIJESA.2021.87202.1013.
[28] R. Pichandi, K. Murugavel Kulandaivelu, K. Alagar, H.K. Dhevaguru, S. Ganesamoorthy, Performance enhancement of photovoltaic module by integrating eutectic inorganic phase change material, Energy Sources, Part A Recover, Util. Environ. Eff. (2020) 1–18.
[29] A. Hasan, J. Sarwar, H. Alnoman, S. Abdelbaqi, Yearly energy performance of a photovoltaic-phase change material (PV-PCM) system in hot climate, Sol. Energy 146 (2017) 417–429.
[30] Y.S. Indartono, S.D. Prakoso, A. Suwono, I.N. Zaini, B. Fernaldi, Simulation and experimental study on effect of phase change material thickness to reduce temperature of photovoltaic panel, IOP Conf. Ser. Mater. Sci. Eng. 88 (2015), 012049, https://doi.org/10.1088/1757-899X/88/1/012049.
[31] M. Cellura, V. Lo Brano, A. Marvuglia, A Photovoltaic panel coupled with a phase changing material heat storage system in hot climates, in: Proc. 25th Conf. Passiv, . Low Energy Archit., Dublin, Ireland, 2008, pp. 22–24.
[32] N. Savvakis, E. Dialyna, T. Tsoutsos, Investigation of the operational performance and efficiency of an alternative PV+PCM concept, Sol. Energy 211 (2020) 1283–1300.
[33] S. Khanna, K.S. Reddy, T.K. Mallick, Performance analysis of tilted photovoltaic system integrated with phase change material under varying operating conditions, Energy 133 (2017) 887–899.
[34] A. Waqas, J. Jie, L. Xu, Thermal behavior of a PV panel integrated with PCM-filled metallic tubes: an experimental study, J. Renew. Sustain. Energy 9 (2017), 053504, https://doi.org/10.1063/1.4995022.
[35] K.S. Kumar, S. Revanth, D. Sanjeev, P.S. Kumar, P. Surya, Experimental investigation of improving the energy conversion efficiency of PV cell by integrating with PCM, Mater, Today Proc. 37 (2020) 712–716.
[36] S. Khanna, K.S. Reddy, T.K. Mallick, Optimization of finned solar photovoltaic phase change material (finned pv pcm) system, Int. J. Therm. Sci. 130 (2018) 313–322, https://doi.org/10.1016/j.ijthermalsci.2018.04.033.
[37] M.J. Huang, P.C. Eames, B. Norton, Phase change materials for limiting temperature rise in building integrated photovoltaics, Sol. Energy 80 (2006) 1121–1130, https://doi.org/10.1016/j.solener.2005.10.006.
[38] S. Preet, B. Bhushan, T. Mahajan, Experimental investigation of water based photovoltaic/thermal (PV/T) system with and without phase change material (PCM), Sol. Energy 155 (2017) 1104–1120, https://doi.org/10.1016/j.solener.2017.07.040.
[39] M.C. Browne, D. Quigley, H.R. Hard, S. Gilligan, N.C.C. Ribeiro, N. Almeida, S.J. McCormack, Assessing the thermal performance of phase change material in a photovoltaic/thermal system, Energy Proc. 91 (2016) 113–121, https://doi.org/10.1016/j.egypro.2016.06.184.
[40] H.M. Maghrabie, K. Elsaid, E.T. Sayed, A. Radwan, A.G. Abo-Khalil, H. Rezk, M.A. Abdelkareem, A.G. Olabi, Phase change materials based on nanoparticles for enhancing the performance of solar photovoltaic panels: a review, J. Energy Storage 48 (2022), 103937, https://doi.org/10.1016/j.est.2021.103937.
[41] M.R. Safaei, H.R. Goshayeshi, I. Chaer, Solar still efficiency enhancement by using graphene oxide/paraffin nano-PCM, Energies 12 (2019) 2002.
[42] A. Shahsavar, S. Khanmohammadi, A. Karimipour, Beta, M. Goodarzi, A novel comprehensive experimental study concerned synthesizes and prepare liquid paraffin-Fe3O4 mixture to develop models for both thermal conductivity & viscosity: a new approach of GMDH type of neural network, Int. J. Heat Mass Tran. 131 (2019) 432–441.
[43] S. Dong, K.S. Al-Zahrani, S.A. Reda, K. Sharma, M.T. Amin, E. Tag-Eldin, M.M. Youshanlouei, Investigation of thermal performance of a shell and tube latent heat thermal energy storage tank in the presence of different nano-enhanced PCMs, Case Stud. Therm. Eng. 37 (2022), 102280.
[44] M.A. Alazwari, M. Algarni, M.R. Safaei, Effects of various types of nanomaterials on PCM melting process in a thermal energy storage system for solar cooling application using CFD and MCMC methods, Int. J. Heat Mass Tran. 195 (2022), 123204.
[45] H.E. Abdelrahman, M.H. Wahba, H.A. Refaey, M. Moawad, N.S. Berbish, Performance enhancement of photovoltaic cells by changing configuration and using PCM (RT35HC) with nanoparticles Al2O3, Sol. Energy 177 (2019) 665–671, https://doi.org/10.1016/j.solener.2018.11.022.
[46] A.S. Abdelrazik, F.A. Al-Sulaiman, R. Saidur, Numerical investigation of the effects of the nano-enhanced phase change materials on the ther-mal and electrical performance of hybrid PV/thermal systems, Energy Convers. Manag. 205 (2020), 112449.
[47] X. Yang, L. Sun, Y. Yuan, X. Zhao, X. Cao, Experimental investigation on performance comparison of PV/T-PCM system and PV/T system, Renew. Energy 119 (2018) 152–159.
[48] M.J. Huang, P.C. Eames, B. Norton, N.J. Hewitt, Natural convection in an internally finned phase change material heat sink for the thermal management of photovoltaics, Sol. Energy Mater. Sol. Cells 95 (2011) 1598–1603.

[49] R.B.-K. Salem Nijmeh, Bashar Hammad, Mohammad Al-Abed, A technical and economic study of a photovoltaic–phase change material (PV-PCM) system in Jordan, Jordan, J. Mech. Ind. Eng. 14 (2020) 371–379.
[50] M. Sarafraz, M.R. Safaei, A. Leon, I. Tlili, T.A. Alkanhal, Z. Tian, M. Goodarzi, M. Arjomandi, Experimental investigation on thermal performance of a PV/T-PCM (Photovoltaic/Thermal) system cooling with a PCM and nanofluid, Energies 12 (2019) 2572.
[51] Y. Cui, J. Zhu, F. Zhang, Y. Shao, Y. Xue, Current status and future development of hybrid PV/T system with PCM module: 4E (energy, exergy, economic and environmental) assessments, Renew. Sustain. Energy. 158 (2022), 112147.
[52] H. Schenck Jr., P.D. Richardson, Theories of Engineering Experimentation, McGraw-Hill, New York, 1972.
[53] S. Kline, F. McClintock, Describing uncertainties in single-sample experiments, Mech. Eng. 75 (1953) 3–8.
[54] S. Khanna, K.S. Reddy, T.K. Mallick, Performance analysis of tilted photovoltaic system integrated with phase change material under varying operating conditions, Energy 133 (2017) 887–899.